

Department of Mechanical Engineering



CAD- II LAB PROJECT

Project Title: Kinematic and Motion Analysis of Car's Windshield Wiper Mechanism

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Abstract

This report presents the kinematic and motion analysis of a windshield wiper mechanism designed in SolidWorks. The study focuses on three assembly configurations small, medium, and large simulating wiper mechanisms for different car sizes. The crank angle vs. wiper angle was recorded for small and medium assemblies to determine the sweep angle. Additionally, motion simulations were conducted to analyze tip speed, crank speed, the wiper path, rocker arm motion, and the linear motion of the slider. The project aims to evaluate the kinematic performance and efficiency of the wiper mechanism under varying geometrical configurations.

Introduction

Windshield wipers are essential components in automotive systems to ensure visibility during rain or snow. The design of a wiper system involves complex linkage mechanisms to achieve the desired wiping pattern. This project involves the kinematic and dynamic analysis of a four-bar wiper linkage using SolidWorks Motion Study. By varying the configuration sizes, the impact of linkage dimensions on the sweep angle and tip velocity was analyzed for small and medium vehicles. This study aids in optimizing the design for better performance and reliability.

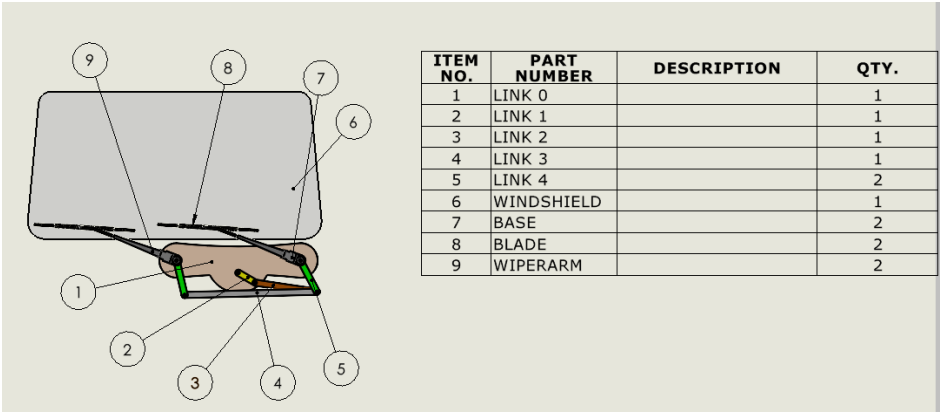


Figure 1:Components of Windshield Wiper Mechanism

Component Dimensions

Table 1: Linkages Dimensions

Component	Small (0.7×)	Medium (1.0×)	Large (1.3×)
Link 0 (Fixed frame)	350 mm	500 mm	650 mm
Link 1 (Crank)	56 mm	80 mm	104 mm
Link 2 (Connecting rod)	175 mm	250 mm	325 mm
Link 3 (Rocker arm)	364 mm	520 mm	676 mm
Link 4 (Wiper arm 1)	99.4 mm	142 mm	184.6 mm
Link 5 (Wiper arm 2)	99.4 mm	142 mm	184.6 mm

Table 2: Wiper and Windshield Dimensions

Component	Small (0.7×)	Medium (1.0×)	Large (1.3×)
Wiper Blade	322 mm	460 mm	598 mm
Wiper Blade Holder	255.5 mm	365 mm	474.5 mm
Windshield Width (Bottom)	840 mm	1200 mm	1560 mm
Windshield Width (Top)	700 mm	1000 mm	1300 mm
Windshield Height	455 mm	650 mm	845 mm

Procedure

1. Model Creation in SolidWorks:

- Designed a wiper mechanism using a four-bar linkage.
- Created three configurations to represent small, medium, and large vehicles.
- Linkage lengths were defined for each configuration.

2. Wiper and Windshield Dimensions Setup:

- Defined wiper blades and holder dimensions.
- Specified windshield dimensions for each configuration.

3. Kinematic Analysis:

- Conducted using SolidWorks Motion Study.
- Rotated crank (Link 1) from 0° to 360° in 30° increments.
- Recorded corresponding wiper arm (Link 4) angle.
- Calculated maximum sweep angle for each configuration.

4. Motion Analysis:

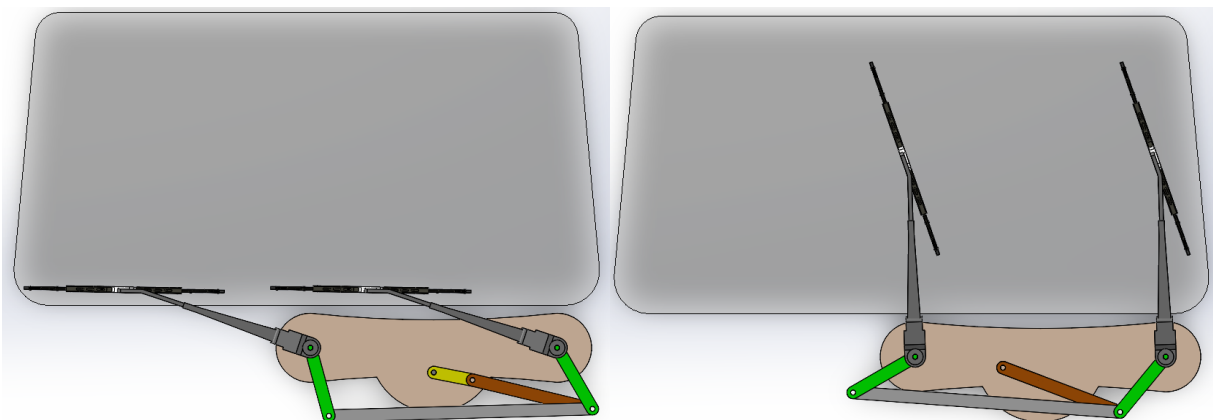
- Simulated crank-driven motion to obtain:
 - Tip speed vs. crank speed.
 - Wiper path trajectory.
 - Rocker arm angular motion.
 - Connecting rod behavior.
 - Linear motion of initial slider mechanism.

Observation

Table 3: θ_2 Vs θ_4 (Medium Car)

θ_2 (Crank Angle)	θ_4 (Wiper Angle)
0°	2.88°
30°	17.91°
60°	39.11°
90°	55.71°
120°	66.32°
150°	71.18°
180°	69.97°
210°	62.2°
240°	48.42°
270°	31.93°
300°	15.7°
330°	3.77°
360°	2.88°

- Max angle: 71.5°
- Sweep Angle = $71.5^\circ - 1.5^\circ \approx 70^\circ$
- Max Tip Velocity of Wiper = **4349 mm/s**
- Max Tip Velocity of Rocker = **1057 mm/s**



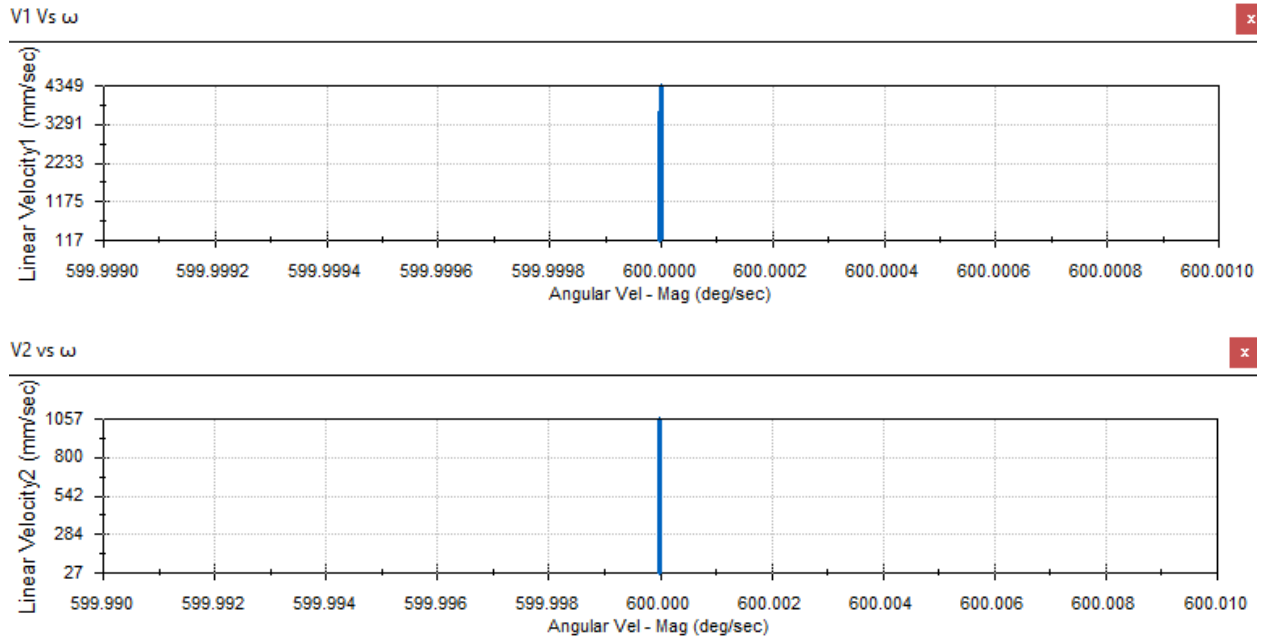


Figure 2,3: Plot of Different Velocity vs Input Velocity (Medium Car)

Table 4: θ_2 Vs θ_4 (Small Car)

θ_2 (Crank Angle)	θ_4 (Wiper Angle)
0°	2.53°
30°	17.54°
60°	38.62°
90°	55.18°
120°	65.6°
150°	70.38°
180°	69.11°
210°	61.17°
240°	47.63°
270°	31.24°
300°	15.12°

330°	3.31°
360°	2.53°

- Max angle: 70.26°
- Sweep Angle = $70.26^\circ - 2.53^\circ \approx 67.7^\circ$
- Max Tip Velocity of Wiper = **1872 mm/s**
- Max Tip Velocity of Rocker = **454 mm/s**

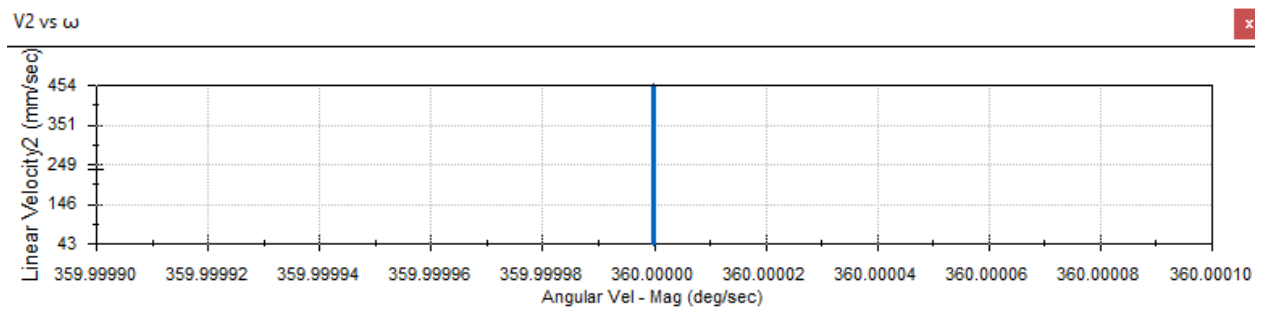
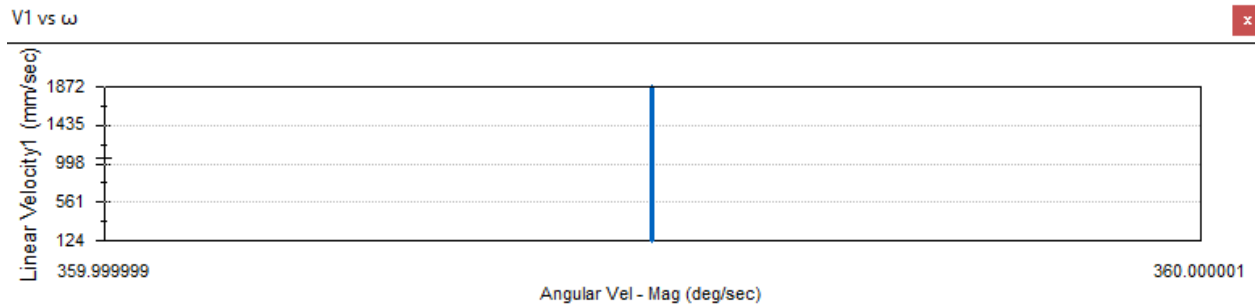


Figure4,5: Plot of Different Velocity vs Input Velocity

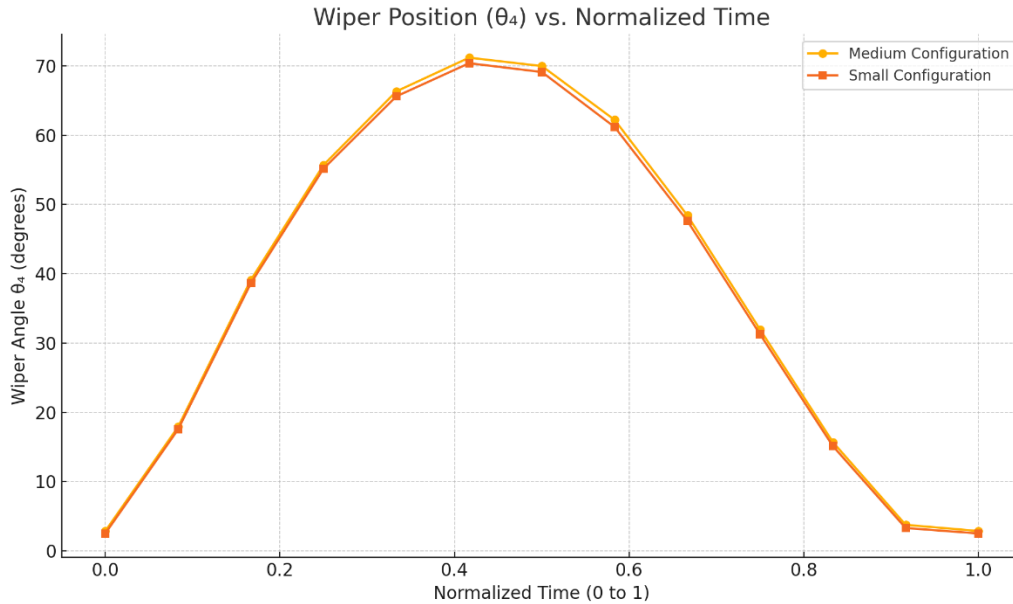


Figure6: Wiper angle vs time

Discussion

The analysis indicates that the medium configuration provides a slightly higher sweep angle compared to the small configuration. This results in improved windshield coverage, making it more suitable for larger vehicles. The crank angle vs. wiper angle relationship demonstrates typical four-bar linkage behavior, with non-linear but smooth angular progression.

Motion simulation confirms that tip speed is directly proportional to crank speed. The connecting rod and rocker arm exhibit proper oscillatory motion, with the rocker showing symmetric angular displacement. The starting slider behaves linearly, confirming the feasibility of crank-driven reciprocating motion.

The **mechanism toggles** (i.e., changes direction and needs more power) when:

- The crank reaches around **90° or 270°**.
- These are the **dead-center positions**, where the connecting rod and rocker are almost in a straight line. More torque is needed here to continue the motion.

These observations support the efficiency and reliability of the linkage design. Scaling dimensions proportionally allows for adapting the mechanism for different vehicle categories while maintaining performance.

Conclusion

The SolidWorks-based simulation of the windshield wiper mechanism has demonstrated effective performance across both small and medium configurations. The sweep angles obtained (approximately 70° and 67.7°) fall within typical design standards. The motion analysis confirmed smooth operation of all linkages, and dynamic results were consistent with expected physical behavior. The study proves that four-bar linkages can be accurately modeled and optimized for different vehicle categories using CAD and motion simulation tools.

References

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